

P2 REVSHOP

1. Introduction

RevShop is a full-stack monolithic e-commerce web application designed to provide an integrated platform for online buying and selling of products.

The application allows two major types of users – Buyers and Sellers – to interact through a centralized system where buyers can explore products, add items to a shopping cart, place orders, and review products, while sellers can manage their inventory, add new products, and monitor orders placed by customers.

The system aims to simulate a real-world online marketplace where multiple users interact with the platform simultaneously. Buyers are able to browse product categories, search for specific items, view product descriptions, and make purchase decisions based on reviews and ratings.

Sellers, on the other hand, can manage product information including product name, description, category, quantity, and pricing.

RevShop is built using modern enterprise technologies including Java, Spring Boot, Spring Security, Oracle SQL, Maven, Log4J, and JUnit. The backend is developed using Spring Boot and follows the MVC architectural pattern, while the frontend uses HTML, CSS, JavaScript, Bootstrap, and Thymeleaf for rendering dynamic content.

The main goal of this project is to demonstrate the development of a complete e-commerce workflow using modern web technologies while ensuring secure authentication, role-based access control, and efficient database management.

2. Objectives of the Project

The main objective of the RevShop project is to design and implement a complete e-commerce web application that allows buyers and sellers to interact through a unified online platform. The system aims to simplify the process of browsing, purchasing, and managing products by providing an intuitive interface and efficient backend services. RevShop enables buyers to explore products, manage their shopping carts, place orders, and review items, while sellers can manage inventory, monitor product performance, and handle customer orders efficiently.

Another important objective of the project is to demonstrate how modern enterprise technologies can be used together to develop a scalable and secure web application. By using technologies such as Java, Spring Boot, Oracle SQL, and Spring Security, the system ensures proper handling of business logic, database management, and user authentication.

The project also aims to implement a structured software architecture that separates different components of the system into layers such as controllers, services, and repositories. This layered architecture improves code maintainability, scalability, and ease of future development.

Additionally, the project focuses on implementing key e-commerce functionalities such as product management, cart management, order processing, payment simulation, and review systems. These features help replicate real-world online shopping workflows and provide users with a complete digital shopping experience.

The specific objectives of the project include:

- To develop a full-stack web application using **Spring Boot** for backend development and **Oracle SQL** for database management.
- To implement **secure user authentication and role-based access control** using Spring Security in order to protect application resources and ensure that only authorized users can access certain functionalities.

- To provide buyers with a convenient and user-friendly platform where they can **browse products, search items, view product details, and purchase products online.**
- To enable sellers to **add, update, and manage product listings**, track inventory levels, and monitor orders placed by customers.
- To implement a **shopping cart system** that allows users to add multiple products, update quantities, and review their selections before proceeding to checkout.
- To simulate **payment processing methods** such as Cash on Delivery, Credit Card, and Debit Card in order to demonstrate the order completion process.
- To allow customers to **submit product reviews and ratings**, helping other users make informed purchasing decisions.
- To demonstrate **enterprise application development practices**, including the use of MVC architecture, layered design, dependency injection, and modular coding.
- To implement **unit testing using JUnit** in order to validate application components and ensure system reliability.
- To manage project dependencies, builds, and configurations efficiently using **Maven.**
- To design a structured and scalable database schema that supports efficient storage and retrieval of application data such as users, products, orders, and reviews.
- To provide a responsive and user-friendly interface using **HTML, CSS, Bootstrap, and Thymeleaf**, ensuring smooth interaction between the frontend and backend components.

3. Scope of the Project

The RevShop system provides an online environment where buyers and sellers interact through a centralized e-commerce platform.

Buyer functionalities include:

- User registration and login
- Product browsing and searching
- Viewing product details such as price, description, and quantity
- Adding products to a shopping cart
- Updating cart quantities
- Removing items from the cart
- Placing orders and making payments
- Viewing order history
- Reviewing purchased products
- Saving favourite products

Seller functionalities include:

- Seller registration and login
- Adding new products
- Updating product information
- Deleting products from inventory
- Managing product quantities and stock levels
- Viewing orders placed for their products
- Monitoring customer reviews and ratings

The system is designed primarily for demonstration and educational purposes but reflects the structure of real-world e-commerce applications.

4. Technology Stack

Backend Technologies:

Java – The primary programming language used for implementing application logic.

Spring Boot – Used for developing REST controllers, business services, and backend components.

Spring Security – Provides authentication, authorization, and role-based access control.

Hibernate / JPA – Handles object-relational mapping between Java classes and database tables.

Frontend Technologies:

HTML – Structures the web pages.

CSS – Provides styling and layout.

Bootstrap – Improves the responsiveness and design of the interface.

JavaScript – Adds interactivity to the application.

Thymeleaf – Used as the server-side template engine for dynamic content rendering.

Database:

Oracle SQL – Used to store application data such as users, products, orders, and reviews.

Testing:

JUnit – Used for unit testing backend components.

Build Tool:

Maven – Used to manage dependencies and project builds.

Logging:

Log4J – Used to track application events and debugging information.

5. System Architecture

RevShop follows a **three-tier architecture**, which separates the application into three independent layers: the **Presentation Layer**, **Application Layer**, and **Database Layer**. This architectural design helps in organizing the system efficiently by separating the user interface, business logic, and data management components. Such separation improves the maintainability, scalability, and security of the system while making the application easier to develop and extend in the future.

The three-tier architecture ensures that each layer performs a specific set of responsibilities and communicates with the other layers through well-defined interfaces. When a user interacts with the system, the request flows through these layers sequentially until the required data is processed and returned to the user interface.

The three main layers of the RevShop architecture are explained below.

5.1 Presentation Layer (User Interface)

The **Presentation Layer** is the topmost layer of the RevShop system where users interact with the application through a web interface. This layer is responsible for displaying information to the user and collecting input from them.

In the RevShop platform, both **buyers and sellers** interact with the application through web pages provided in this layer. Buyers can browse products, view product details, manage their cart, and place orders. Sellers can manage their product inventory, view orders, and update product information.

The presentation layer includes various user interface components such as:

- Login and registration pages
- Product listing pages
- Product detail pages
- Shopping cart pages
- Checkout and payment pages
- Seller dashboard and product management pages

Technologies used in this layer include:

- **HTML** for structuring the web pages
- **CSS** for styling and layout
- **Bootstrap** for responsive design and UI components
- **JavaScript** for interactive functionality
- **Thymeleaf** as the server-side template engine for rendering dynamic content

This layer sends user requests to the application layer and displays the response returned by the system.

5.2 Application Layer (Business Logic)

The **Application Layer** is the core processing layer of the RevShop system. It handles the business logic and processes requests received from the presentation layer.

This layer is responsible for validating user inputs, executing application logic, managing transactions, and communicating with the database layer to retrieve or store data.

The application layer consists of several components including:

Controllers

Controllers act as the entry point for user requests. They receive HTTP requests from the presentation layer and forward them to the appropriate service classes for processing.

Examples of controllers in RevShop include:

- Buyer Controller
- Seller Controller
- Product Controller
- Cart Controller
- Order Controller
- Payment Controller

Services

The **service layer** contains the business logic of the system. It processes operations such as product management, cart updates, order creation, and payment processing.

Services interact with repository classes to perform database operations.

Security Configuration

RevShop uses **Spring Security** to implement authentication and authorization mechanisms. This ensures that only authorized users can access protected features of the application.

Security features include:

- User login authentication
- Role-based access control for buyers and sellers
- Password encryption
- Secure session handling

Technologies used in the application layer include:

- **Spring Boot**
 - **Spring Security**
 - **Spring Data JPA / Hibernate**
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5.3 Database Layer (Data Storage)

The **Database Layer** is responsible for storing and managing all persistent data used by the RevShop application. This layer ensures that user data, product information, and transaction records are stored securely and retrieved efficiently when needed.

The database used in RevShop is **Oracle SQL**, which is a powerful relational database management system capable of handling complex queries and large datasets.

The database stores various types of information including:

- User accounts and profile details
- Buyer and seller information
- Product details such as name, price, description, and quantity
- Shopping cart information
- Order and payment records
- Product reviews and ratings
- Notification records

The application communicates with the database using **Spring Data JPA and Hibernate**, which provide object-relational mapping (ORM) capabilities. ORM allows Java objects to be mapped directly to database tables, reducing the need for complex SQL queries and simplifying database operations.

5.4 Data Flow in the System

The interaction between these three layers follows a structured workflow:

1. The user interacts with the **presentation layer** through the web interface.
2. The user request is sent to the **application layer**, where controllers process the request and invoke the required services.
3. The service layer communicates with the **database layer** to retrieve or store data.
4. The database returns the requested data to the application layer.
5. The processed response is returned to the presentation layer and displayed to the user.

This layered architecture ensures a clear separation between the user interface, application logic, and data storage components.

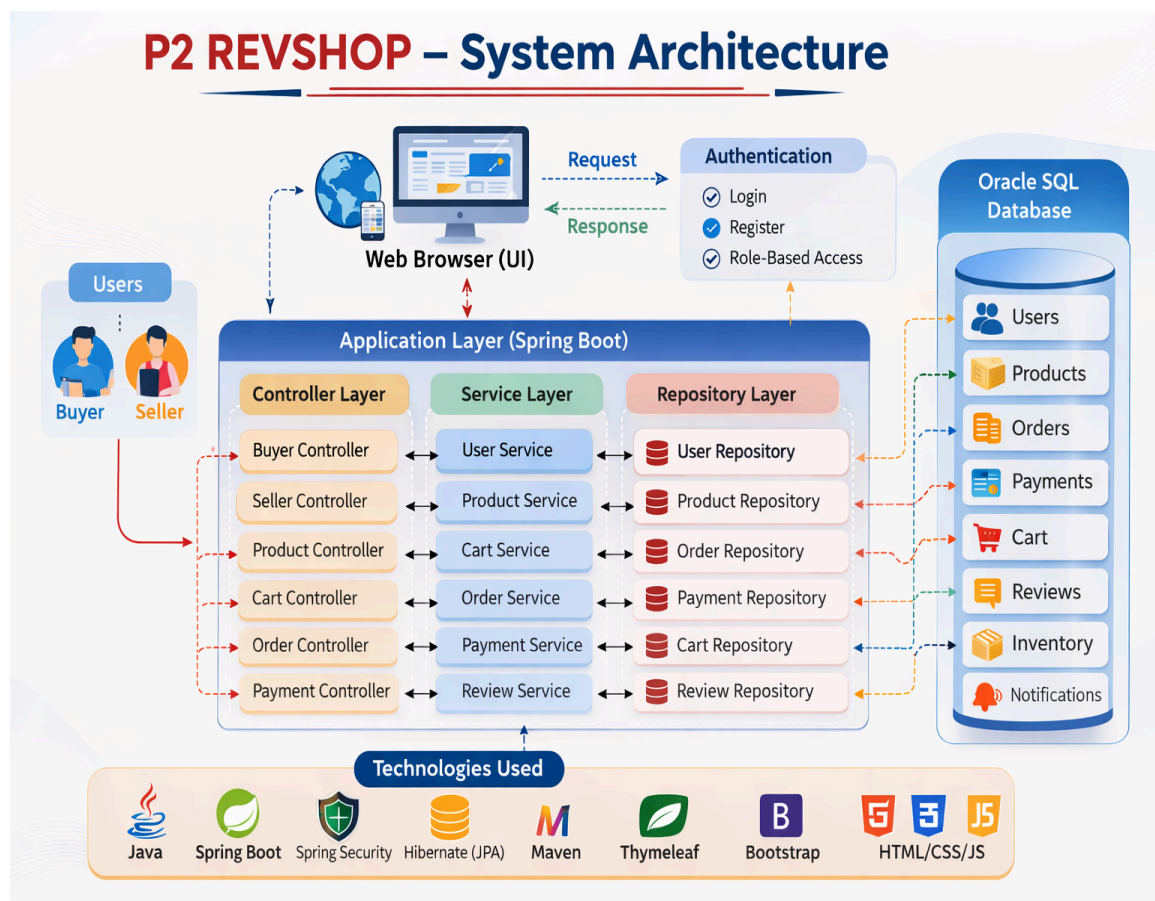
5.5 Benefits of Three-Tier Architecture

The three-tier architecture used in RevShop provides several advantages:

- **Improved maintainability** by separating responsibilities into different layers
- **Better scalability** since each layer can be modified or expanded independently
- **Enhanced security** by restricting direct database access

- **Simplified development and testing** due to modular design
- **Better code organization** and easier debugging

This architectural approach allows the RevShop system to remain flexible, maintainable, and ready for future enhancements.



6. Project Structure

The project follows a modular Spring Boot project structure.

Important folders include:

controller – Handles incoming HTTP requests.

service – Contains business logic.

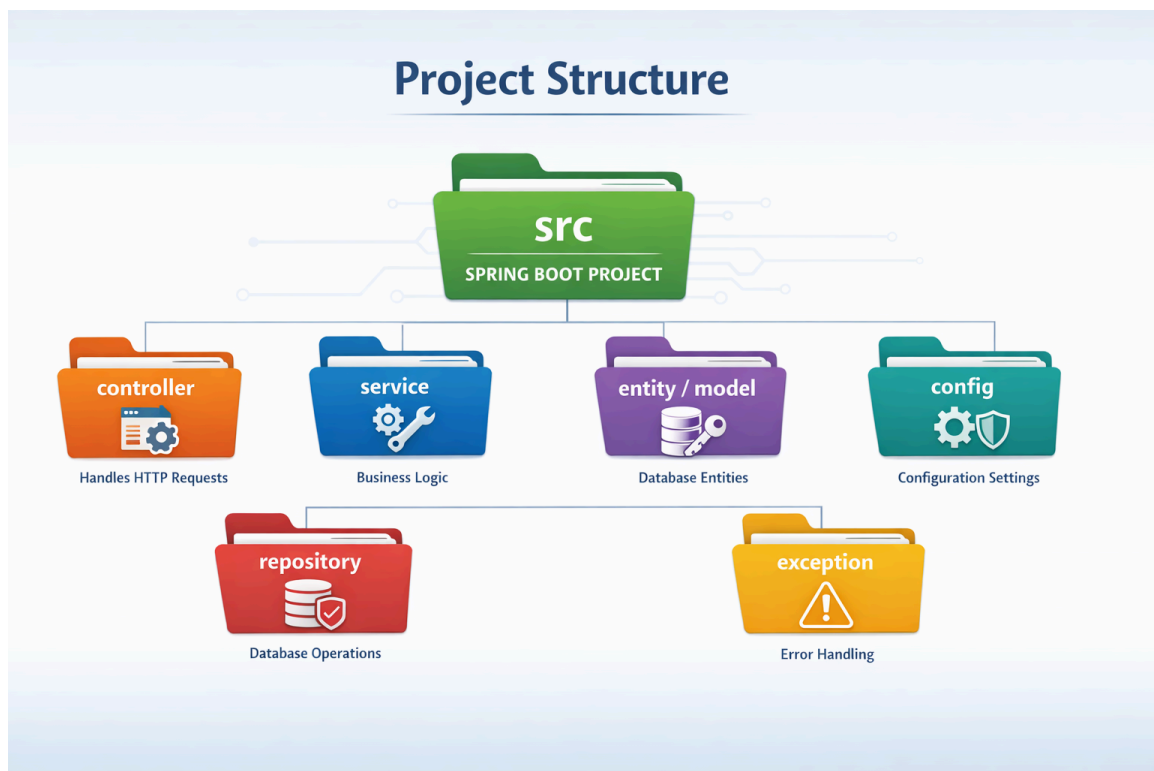
repository – Manages database operations.

entity/model – Defines database entities.

config – Contains configuration classes such as security settings.

exception – Handles custom application exceptions.

This structured approach ensures that the application remains organized and easy to maintain.



7. User Roles

RevShop supports two primary user roles.

Buyer:

Buyers can browse products, search for items, add products to their cart, place orders, review products, and manage their favourite items.

Seller:

Sellers can add new products, update product details, manage inventory levels, view orders placed by customers, and monitor product reviews.

8. Functional Modules

The RevShop application is divided into several functional modules, each responsible for performing a specific set of operations within the system. These modules work together to ensure smooth interaction between buyers and sellers while maintaining efficient management of products, orders, and user information.

By dividing the system into modules, the application becomes easier to maintain, test, and expand in the future. Each module focuses on a specific functionality such as authentication, product management, cart operations, order processing, payment simulation, and notifications.

The major functional modules of the RevShop system are described below.

8.1 Authentication and User Management Module

The Authentication and User Management module is responsible for handling user registration, login, and account management. This module ensures that only authorized users can access the system.

Users must register by providing personal details such as name, email address, password, phone number, and address. Once registered, users can log in using their credentials.

Spring Security is used to manage authentication and authorization. Passwords are encrypted before being stored in the database to ensure data security.

Key features of this module include:

- User registration
- Secure login and logout
- Role-based access control (Buyer and Seller roles)
- Profile management
- Password security and encryption

This module plays an important role in protecting system data and ensuring that only authorized users can perform specific actions.

8.2 Product Management Module

The Product Management module allows sellers to manage products available in the RevShop platform. Sellers can add new products, update product information, and remove products from their inventory.

Each product contains details such as product name, description, category, price, and available quantity. Sellers can update these details whenever required.

This module also allows buyers to browse products and view product details before making a purchase.

Key features include:

- Adding new products
- Updating product information
- Deleting products from inventory
- Viewing product listings
- Managing product categories
- Displaying product details to buyers

This module ensures that the product catalog remains updated and accessible to buyers.

8.3 Product Browsing and Search Module

The Product Browsing and Search module allows buyers to explore products available on the platform. Users can browse products by category or search for specific products using keywords.

This module improves the shopping experience by allowing users to easily find products that match their interests.

Features of this module include:

- Viewing product listings
- Searching products using keywords
- Filtering products by category
- Viewing product details such as description, price, and availability

The search functionality helps buyers quickly locate products within a large product catalog.

8.4 Cart Management Module

The Cart Management module allows buyers to manage the products they intend to purchase. Buyers can add products to their cart and update quantities before placing an order.

The cart acts as a temporary storage area where users can review selected products before completing the purchase.

Features of the cart module include:

- Adding products to the shopping cart
- Updating product quantities
- Removing products from the cart
- Viewing cart summary with total price calculation

This module ensures that buyers can manage their product selections conveniently before proceeding to checkout.

8.5 Order Management Module

The Order Management module is responsible for processing customer orders and maintaining order records. Once a buyer completes the checkout process, an order is created and stored in the database.

Both buyers and sellers can view order information through this module.

Buyers can track their order history and view order details, while sellers can monitor orders placed for their products.

Key features include:

- Placing orders
- Viewing order details
- Tracking order history
- Displaying order information for sellers

This module ensures proper handling and tracking of customer orders.

8.6 Payment Module

The Payment module handles the payment process during checkout. In RevShop, the payment system is simulated to demonstrate the order completion process.

Users can choose from different payment options such as:

- Cash on Delivery
- Credit Card
- Debit Card

Payment information is recorded in the database and linked to the corresponding order.

This module demonstrates how payment processing works in an e-commerce application.

8.7 Review and Rating Module

The Review and Rating module allows buyers to share feedback about products they have purchased. Customers can submit reviews and rate products based on their experience.

These reviews help other buyers make informed purchasing decisions.

Features of this module include:

- Submitting product reviews
- Providing product ratings
- Viewing reviews from other customers

This module improves transparency and trust within the platform.

8.8 Notification Module

The Notification module provides in-app notifications to inform users about important system events. Notifications help users stay updated about activities such as order confirmations and product updates.

Examples of notifications include:

- Order placed confirmation for buyers
- New order notification for sellers
- Low stock alerts for sellers

This module enhances user engagement and ensures timely communication within the system.

8.9 Inventory Management Module

The Inventory Management module allows sellers to track product stock levels and manage product availability. Sellers can update product quantities whenever new stock arrives or when products are sold.

Features include:

- Updating product stock levels
- Monitoring inventory availability
- Receiving alerts when stock levels are low

This module ensures that product availability is accurately reflected in the system.

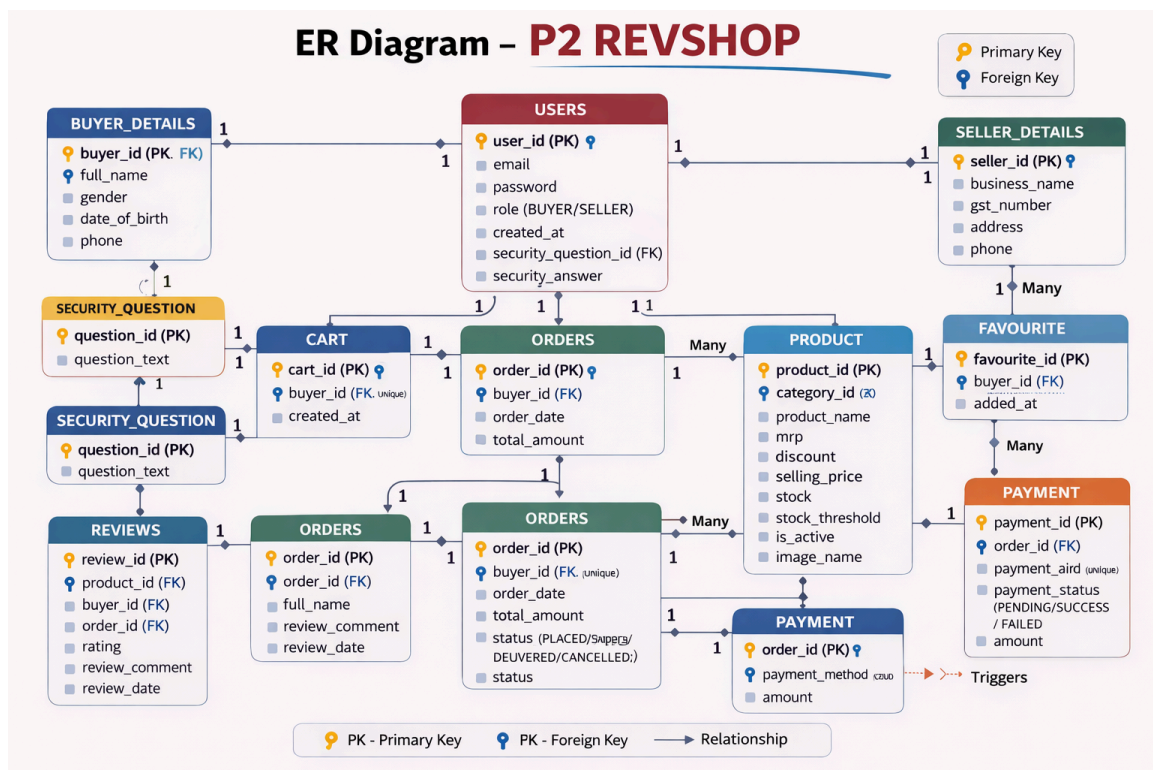
9. Database Design

The RevShop database stores all application data.

Key entities include:

Users
BuyerDetails
SellerDetails
Product
Category
Cart
CartItem
Order
OrderItem
Payment
Review
Favourite
Notification

Relationships between these tables allow the system to manage user interactions and transactions efficiently.



10. Implementation

The implementation of RevShop focuses on building a scalable backend system with a responsive frontend interface.

Spring Boot controllers handle HTTP requests and communicate with service classes that contain business logic.

The service layer interacts with repository classes which manage database operations using Spring Data JPA.

The frontend uses Thymeleaf templates to display dynamic data received from the backend.

Bootstrap is used to create responsive layouts while JavaScript provides interactive features.

11. Testing

Testing ensures that the RevShop system functions correctly and reliably.

Unit Testing:

JUnit is used to test individual service methods and controller endpoints.

Integration Testing:

Integration testing verifies interactions between controllers, services, repositories, and the database.

User Testing:

User testing verifies real-world workflows such as user registration, product browsing, cart management, and order placement.

12. Results and Discussion

The RevShop application successfully demonstrates the implementation of an e-commerce platform with both buyer and seller functionality.

Buyers are able to browse products, manage carts, and place orders efficiently.

Sellers are able to manage inventory and monitor orders placed by customers.

The system integrates authentication, database management, and frontend interfaces to provide a complete online shopping experience.

13. Advantages of the System

- Secure authentication and authorization using Spring Security.
- Structured architecture using MVC design.
- Efficient product and inventory management.
- Interactive user interface using Bootstrap.
- Scalable backend system using Spring Boot.
- Reliable database management using Oracle SQL.

14. Future Enhancements

Future improvements for RevShop may include:

- Integration with real payment gateways such as Stripe or Razorpay.
- Mobile application development.
- AI-based product recommendation systems.
- Real-time order tracking.
- Advanced analytics dashboards for sellers.

15. Conclusion

RevShop demonstrates how a full-stack e-commerce application can be developed using modern enterprise technologies.

The project integrates backend services, database management, and responsive user interfaces to deliver a complete online shopping platform.

By implementing secure authentication, product management, cart functionality, and order processing, the system provides a strong foundation for a scalable e-commerce solution.